BTS-I(R/S)/II(S)-03.22-0059 Reg. No. | | | | | | | | |

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B.Tech. Degree I Semester Regular/Supplementary and II Semester Supplementary Examination March 2022

CE/EE/ME/SE 19-200-0104A & CS/EC/IT 19-200-0204B
BASIC ELECTRICAL ENGINEERING
(*2019 Scheme*)

Time: 3 Hours Maximum Marks: 60

PART A
(Answer *ALL* questions)

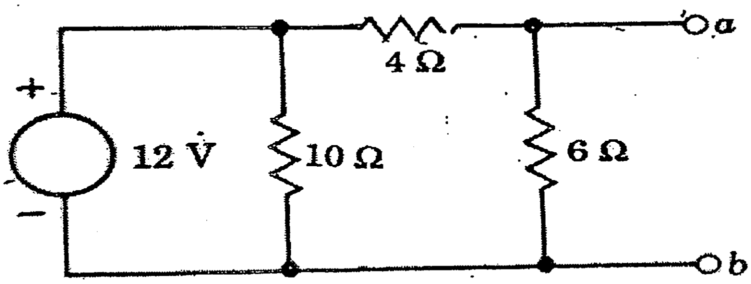
($8 \times 3 = 24$)

I. (a) State Kirchhoff's laws.
(b) Explain Thevenin's law of electric circuits.
(c) Write similarities and dissimilarities of electric and magnetic circuit.
(d) A mild steel ring having cross-sectional area 500 mm^2 and mean circumference 400mm has a coil of 200 turns wound uniformly around it.
Calculate:
(i) Reluctance
(ii) Current required to produce a flux of 800 µwb.
Take relative permeability of mild steel as 2000.
(e) Discuss RMS value and Form Factor of sinusoidal alternating waveform.
(f) Explain series RLC circuit resonance condition and derive the resonant frequency.
(g) Explain the different levels of power transmission by suitable illustration.
(h) Discuss the working principle of Transformer.

PART B

($4 \times 12 = 48$)

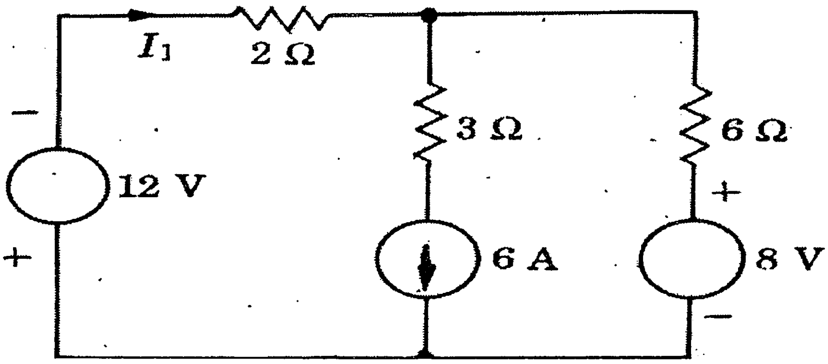
II. For the circuit shown in figure draw the Norton's equivalent circuit at terminals *a, b*. (12)



OR

(P.T.O.)

III.		Determine the current I_1 in the circuit given below by superposition theorem.	(12)



IV.		Explain the working principle of Energy meter with neat diagram.	(12)

OR

V.	(a)	Define: (i) MMF (ii) Magnetizing force (iii) Flux density (iv) Reluctance	(6)
	(b)	State and explain Faraday's Law of Electromagnetic Induction.	(6)
VI.		A coil of resistance 100 Ω and inductance 100 μH is connected in series with a 100 pF capacitor. The circuit is connected to a 10 V variable frequency source. Calculate: (i) The resonant frequency (ii) Current at resonance (iii) Voltage across L & C at resonance. (iv) Q-factor of the circuit.	(12)

OR

VII.	(a)	Derive the relation between phase voltage and line voltage of star connected three phase system. Include phasor diagram.	(8)
	(b)	The apparent power drawn by an ac circuit is 10 kVA and active power is 8 kW. What is the reactive power in the circuit? What is the power factor of the circuit?	(4)
VIII.		With neat sketches explain the working of a Thermal power plant.	(12)

OR

IX.	(a)	Explain the principle of energy generation from solar energy.	(6)
	(b)	With a neat sketch describe the constructional details of DC machine.	(6)

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